

Advanced Manual **Smart Contract Audit**

September 1, 2026

 [CoinsultAudits](https://twitter.com/CoinsultAudits)

 t.me/coinsult_tg

 coinsult.net

Audit requested by



Vortex

0x04E51527200b1f8A24D8696765Da7A2216265ec3

Global Overview

Manual Code Review

In this audit report we will highlight the following issues:

Vulnerability Level	Total	Pending	Acknowledged	Resolved
● Informational	0	0	0	0
● Low-Risk	1	1	0	0
● Medium-Risk	2	2	0	0
● High-Risk	0	0	0	0

Centralization Risks

Coinsult checked the following privileges:

Contract Privilege	Description
Owner needs to enable trading?	● Owner needs to manually enable trading
Owner can mint?	● Owner can mint new tokens
Owner can blacklist?	● Owner cannot blacklist addresses
Owner can set fees?	● Owner can set the sell fee to 0%
Owner can exclude from fees?	● Owner cannot exclude from fees
Can be honeypotted?	● Contract can become honeypot
Owner can set Max TX amount?	● Owner cannot set max transaction amount

More owner privileges are listed later in the report.

Table of Contents

1. Audit Summary

- 1.1 Audit scope
- 1.2 Tokenomics
- 1.3 Source Code

2. Disclaimer

3. Global Overview

- 3.1 Informational issues
- 3.2 Low-risk issues
- 3.3 Medium-risk issues
- 3.4 High-risk issues

4. Vulnerabilities Findings

5. Contract Privileges

- 5.1 Maximum Fee Limit Check
- 5.2 Contract Pausability Check
- 5.3 Max Transaction Amount Check
- 5.4 Exclude From Fees Check
- 5.5 Ability to Mint Check
- 5.6 Ability to Blacklist Check
- 5.7 Owner Privileges Check

6. Notes

- 6.1 Notes by Coinsult
- 6.2 Notes by Vortex

7. Contract Snapshot

8. Website Review

9. Certificate of Proof

Audit Summary

Project Name	Vortex
Website	https://vortexdao.app/
Blockchain	Binance Smart Chain
Smart Contract Language	Solidity
Contract Address	0x04E51527200b1f8A24D8696765Da7A2216265ec3
Audit Method	Static Analysis, Manual Review
Date of Audit	1 September 2026

This audit report has been prepared by Coinsult's experts at the request of the client. In this audit, the results of the static analysis and the manual code review will be presented. The purpose of the audit is to see if the functions work as intended, and to identify potential security issues within the smart contract.

The information in this report should be used to understand the risks associated with the smart contract. This report can be used as a guide for the development team on how the contract could possibly be improved by remediating the issues that were identified.

Audit Scope

Coinsult was commissioned by Vortex to perform an audit based on the following code:

<https://bscscan.com/token/0x04E51527200b1f8A24D8696765Da7A2216265ec3#code>

Note that we only audited the code available to us on this URL at the time of the audit. If the URL is not from any block explorer (main net), it may be subject to change. Always check the contract address on this audit report and compare it to the token you are doing research for.

Audit Method

Coinsult's manual smart contract audit is an extensive methodical examination and analysis of the smart contract's code that is used to interact with the blockchain. This process is conducted to discover errors, issues and security vulnerabilities in the code in order to suggest improvements and ways to fix them.

Automated Vulnerability Check

Coinsult uses software that checks for common vulnerability issues within smart contracts. We use automated tools that scan the contract for security vulnerabilities such as integer-overflow, integer-underflow, out-of-gas-situations, unchecked transfers, etc.

Manual Code Review

Coinsult's manual code review involves a human looking at source code, line by line, to find vulnerabilities. Manual code review helps to clarify the context of coding decisions. Automated tools are faster but they cannot take the developer's intentions and general business logic into consideration.

Used tools

- Slither: Solidity static analysis framework
- Remix: IDE Developer Tool
- CWE: Common Weakness Enumeration
- SWC: Smart Contract Weakness Classification and Test Cases
- DEX: Testnet Blockchains

Risk Classification

Coinsult uses certain vulnerability levels, these indicate how bad a certain issue is. The higher the risk, the more strictly it is recommended to correct the error before using the contract.

Vulnerability Level	Description
● Informational	Does not compromise the functionality of the contract in any way
● Low-Risk	Won't cause any problems, but can be adjusted for improvement
● Medium-Risk	Will likely cause problems and it is recommended to adjust
● High-Risk	Will definitely cause problems, this needs to be adjusted

Coinsult has four statuses that are used for each risk level. Below we explain them briefly.

Risk Status	Description
Total	Total amount of issues within this category
Pending	Risks that have yet to be addressed by the team
Acknowledged	The team is aware of the risks but does not resolve them
Resolved	The team has resolved and remedied the risk

SWC Attack Analysis

The Smart Contract Weakness Classification Registry (SWC Registry) is an implementation of the weakness classification scheme proposed in EIP-1470. It is loosely aligned to the terminologies and structure used in the Common Weakness Enumeration (CWE) while overlaying a wide range of weakness variants that are specific to smart contracts.

ID	Description	Status
SWC-100	Function Default Visibility	Passed
SWC-101	Integer Overflow and Underflow	Passed
SWC-102	Outdated Compiler Version	Passed
SWC-103	Floating Pragma	Passed
SWC-104	Unchecked Call Return Value	Passed
SWC-105	Unprotected Ether Withdrawal	Passed
SWC-106	Unprotected SELFDESTRUCT Instruction	Passed
SWC-107	Reentrancy	Passed
SWC-108	State Variable Default Visibility	Passed
SWC-109	Uninitialized Storage Pointer	Passed
SWC-110	Assert Violation	Passed
SWC-111	Use of Deprecated Solidity Functions	Passed
SWC-112	Delegatecall to Untrusted Callee	Passed
SWC-113	DoS with Failed Call	Passed
SWC-114	Transaction Order Dependence	Passed
SWC-115	Authorization through tx.origin	Passed

SWC-116	Block values as a proxy for time	Passed
SWC-117	Signature Malleability	Passed
SWC-118	Incorrect Constructor Name	Passed
SWC-119	Shadowing State Variables	Passed
SWC-120	Weak Sources of Randomness from Chain Attributes	Passed
SWC-121	Missing Protection against Signature Replay Attacks	Passed
SWC-122	Lack of Proper Signature Verification	Passed
SWC-123	Requirement Violation	Passed
SWC-124	Write to Arbitrary Storage Location	Passed
SWC-125	Incorrect Inheritance Order	Passed
SWC-126	Insufficient Gas Griefing	Passed
SWC-127	Arbitrary Jump with Function Type Variable	Passed
SWC-128	DoS With Block Gas Limit	Passed
SWC-129	Typographical Error	Passed
SWC-130	Right-To-Left-Override control character (U+202E)	Passed
SWC-131	Presence of unused variables	Passed
SWC-132	Unexpected Ether balance	Passed
SWC-133	Hash Collisions With Multiple Variable Length Arguments	Passed
SWC-134	Message call with hardcoded gas amount	Passed
SWC-135	Code With No Effects	Passed
SWC-136	Unencrypted Private Data On-Chain	Passed

Error Code	Description
CS-01	Wrong revert string in

● **Low-Risk:** Could be fixed, will not bring problems.

Wrong revert string in

```
function addwhitelisted(address _whitelisted,bool _status) external onlyAdmin {  
    require(_whitelisted != address(0x0) ,"both address are same");  
    whitelisted[_whitelisted] = _status ;  
  
}
```

Recommendation

Should be something about _whitelisted address cannot be the zero address

Error Code	Description
CSM-01	RenounceOwnership is misleading as no function uses the onlyOwner modifier but is authorized via modifier onlyAdmin() and modifier onlyWhite()

● **Medium-Risk:** Should be fixed, could bring problems.

RenounceOwnership is misleading as no function uses the onlyOwner modifier but is authorized via modifier onlyAdmin() and modifier onlyWhite()

```
function renounceOwnership() public onlyOwner {
    emit OwnershipTransferred(_owner, address(0));
    _owner = address(0);
}
```

Recommendation

Make sure functions depend on onlyOwner modifier or remove RenounceOwnership function.

Error Code	Description
CSM-02	<code>_user</code> is useless here as there is only one address (deployer address) for which tokens can be unlocked for

● **Medium-Risk:** Should be fixed, could bring problems.

`_user` is useless here as there is only one address (deployer address) for which tokens can be unlocked for

```
function unlocktoken(address _user,uint256 _amount) external onlyAdmin {  
    require(premintlock[_user] >= _amount,"enter valid amount");  
    premintlock[_user] -= _amount ;  
    emit premintunlock(_user,_amount,block.timestamp) ;  
}
```

Recommendation

Remove the `_user` parameter to avoid failing intended actions.

Maximum Fee Limit Check

Error Code	Description
CEN-01	Centralization: Operator Fee Manipulation

Coinsult tests if the owner of the smart contract can set the transfer, buy or sell fee to 25% or more. It is bad practice to set the fees to 25% or more, because owners can prevent healthy trading or even stop trading when the fees are set too high.

Type of fee	Description
Max transfer fee	0%
Max buy fee	0%
Max sell fee	0%

Contract Honeypot Check

Error Code	Description
CEN-02	Centralization: Operator Pausability

Coinsult tests if the owner of the smart contract has the ability to pause the contract. If this is the case, users can no longer interact with the smart contract; users can no longer trade the token.

Privilege Check	Description
Can owner pause the contract?	● Owner can pause the smart contract

Function

```
function pauseTrading() external onlyAdmin {
    tradingPaused = true;
    emit TradingPaused(true, block.timestamp);
}
```

Max Transaction Amount Check

Error Code	Description
CEN-03	Centralization: Operator Transaction Manipulation

Coinsult tests if the owner of the smart contract can set the maximum amount of a transaction. If the transaction exceeds this limit, the transaction will revert. Owners could prevent normal transactions to take place if they abuse this function.

Privilege Check	Description
Can owner set max tx amount?	 Owner cannot set max transaction amount

Exclude From Fees Check

Error Code	Description
CEN-04	Centralization: Operator Exclusion

Coinsult tests if the owner of the smart contract can exclude addresses from paying tax fees. If the owner of the smart contract can exclude from fees, they could set high tax fees and exclude themselves from fees and benefit from 0% trading fees. However, some smart contracts require this function to exclude routers, dex, cex or other contracts / wallets from fees.

Privilege Check	Description
Can owner exclude from fees?	● Owner cannot exclude from fees

Ability To Mint Check

Error Code	Description
CEN-05	Centralization: Operator Increase Supply

Coinsult tests if the owner of the smart contract can mint new tokens. If the contract contains a mint function, we refer to the token's total supply as non-fixed, allowing the token owner to "mint" more tokens whenever they want.

A mint function in the smart contract allows minting tokens at a later stage. A method to disable minting can also be added to stop the minting process irreversibly.

Minting tokens is done by sending a transaction that creates new tokens inside of the token smart contract. With the help of the smart contract function, an unlimited number of tokens can be created without spending additional energy or money.

Privilege Check	Description
Can owner mint?	● Owner can mint new tokens

Function

```
function mint(uint256 amount) external onlyWhite nonReentrant returns (bool) {
    _mint(_msgSender(), amount);
    emit mintname(msg.sender, amount, block.timestamp);
    return true;
}
```

Enable Trading

Error Code	Description
CEN-06	Centralization: Operator enable trading

Coinsult tests if the owner of the smart contract needs to manually enable trading before everyone can buy & sell. If the owner needs to manually enable trading, this poses a high centralization risk.

If the owner needs to manually enable trading, make sure to check if the project has a SAFU badge or a trusted KYC badge. Always DYOR when investing in a project that needs to manually enable trading.

Privilege Check	Description
Owner needs to enable trading?	 Owner needs to manually enable trading

Ability To Blacklist Check

Error Code	Description
CEN-07	Centralization: Operator Dissallows Wallets

Coinsult tests if the owner of the smart contract can blacklist accounts from interacting with the smart contract. Blacklisting methods allow the contract owner to enter wallet addresses which are not allowed to interact with the smart contract.

This method can be abused by token owners to prevent certain / all holders from trading the token. However, blacklists might be good for tokens that want to rule out certain addresses from interacting with a smart contract.

Privilege Check	Description
Can owner blacklist?	● Owner cannot blacklist addresses

Other Owner Privileges Check

Error Code	Description
CEN-100	Centralization: Operator Privileges

Coinsult lists all important contract methods which the owner can interact with.

Owner can add any address to admin address which allows these addresses to pause trading and unlock tokens

Owner can add any address to whitelisted address which allows these addresses to mint tokens

Notes

Notes by Vortex

No notes provided by the team.

Notes by Coinconsult

No notes provided by Coinconsult

Contract Snapshot

This is how the constructor of the contract looked at the time of auditing the smart contract.

```
contract VRX is Context, IBEP20, Ownable, ReentrancyGuard {
    using SafeMath for uint256;

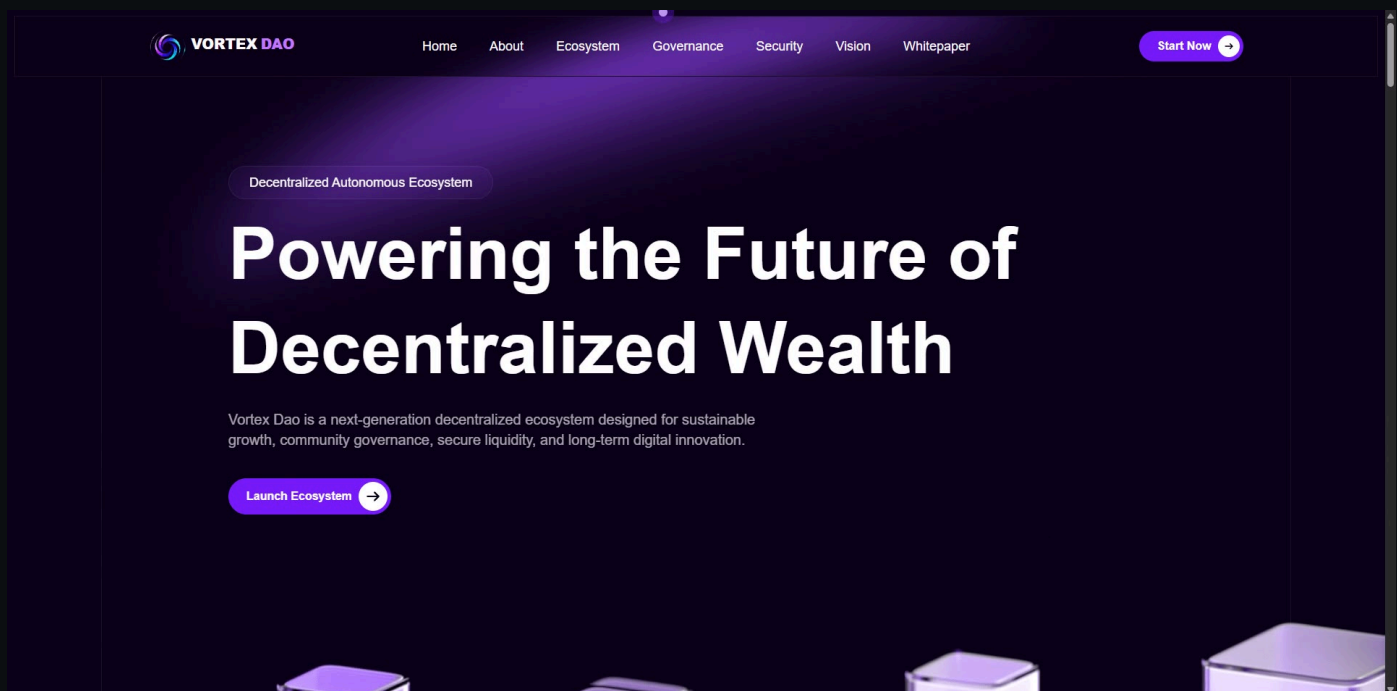
    mapping (address => uint256) private _balances;

    mapping (address => mapping (address => uint256)) private _allowances;

    uint256 circulationSUPPLY ;
    uint256 private _totalSupply;
    uint8 public _decimals;
    string public _symbol;
    string public _name;
```

Website Review

Coinsult checks the website completely manually and looks for visual, technical and textual errors. We also look at the security, speed and accessibility of the website. In short, a complete check to see if the website meets the current standard of the web development industry.



Type of check	Description
Mobile friendly?	● The website is mobile friendly
Contains jQuery errors?	● The website does not contain jQuery errors
Is SSL secured?	● The website is SSL secured
Contains spelling errors?	● The website does not contain spelling errors

Certificate of Proof

● Not KYC verified by Coinsult

Vortex

Audited by Coinsult.net



Date: 1 September 2026

✓ Advanced Manual Smart Contract Audit

Disclaimer

This audit report has been prepared by Coinsult's experts at the request of the client. In this audit, the results of the static analysis and the manual code review will be presented. The purpose of the audit is to see if the functions work as intended, and to identify potential security issues within the smart contract.

The information in this report should be used to understand the risks associated with the smart contract. This report can be used as a guide for the development team on how the contract could possibly be improved by remediating the issues that were identified.

Coinsult is not responsible if a project turns out to be a scam, rug-pull or honeypot. We only provide a detailed analysis for your own research.

Coinsult is not responsible for any financial losses. Nothing in this contract audit is financial advice, please do your own research.

The information provided in this audit is for informational purposes only and should not be considered investment advice. Coinsult does not endorse, recommend, support or suggest to invest in any project.

Coinsult can not be held responsible for when a project turns out to be a rug-pull, honeypot or scam.

End of report

Smart Contract Audit

 CoinsultAudits

 info@coinsult.net

 coinsult.net

Request your smart contract audit / KYC

t.me/coinsult_tg